SEMESTER/YEAR : **III SEM/ II YEAR**
COURSE CODE : **25MBAS001**
TITLE OF THE COURSE : **INTERNATIONAL SUPPLY CHAIN OPERATIONS PLANNING**
L: T: P: C : **3: 0: 2: 4**

Overview

This course has been designed to give the students a holistic view of planning and operations of supply chains. It has been compiled with the users and service providers in international logistics in mind. It covers all the concepts that are important to personnel who are involved in export/import operations. All relevant issues are explained at length, this includes documentation, terms of payment, INCOTERMs, insurance & claims, risk management & currencies, and a lot more.
The concepts need to be understood and so are clearly and accurately portrayed. The vocabulary too needs to be precise. Many of the topics are technical, but need to be known by every logistics professional. Processes too need to be integrated into every business area. This course does just that.

Course Objectives

1. Explain the principles and functions of International Supply Chain Operations Planning (ISCOP)
2. Describe various technical aspects of International Supply Chain Management
3. Develop an understanding of documentation related to international trade, transportation, insurance etc.
4. Summarize various aspects of international terminal operations like customs clearance, storage and handling
5. Practice the elements of data collection, demand and supply planning, pre-executive and executive meetings as part of S&OP process

Course Outcomes

Upon successful completion of this course, the student will have reliably demonstrated ability to

1. Exhibit the fair understanding of concepts, theories, models and principles of International SC management and operations
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of International SC management and operations in real time problem solving
3. Demonstrate the ability of analyzing the real time problems using skill acquired by concepts, theories, models and principles of International SC management and operations

4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of International SC management and operations in real time problem solving course
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of International SC management and operations in real time problem solving course

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Supply Chain practices by way of: ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

<u>**Units**</u>	<u>**Syllabus Details**</u>	<u>**Teaching Hours**</u>
Unit I	**Basic Concepts of International Trade & Markets** ● International Trade Concepts ● The Place of Supply Chains in International Trade ● International Logistics Infrastructure ● Methods of Entry into International Markets	8
Unit II	**International Transport Planning** ● Modes of Transport and the Role of Intermediaries ● Contracts & Agreements in International Trade ● INCOTERMs ● Terms of Payment/Currencies & Managing Transaction Risks ● Introduction to Green Supply Chain Management	8
Unit III	**Documentation, Packaging & Security** ● Trade & Transportation Documents ● Insurance & Risk Management ● Packaging for International Trade ● International Logistics Security	8

Unit IV	**Customs Clearance, Storage & Handling** ● Cross-border Trade ● Customs Clearance ● International Terminal Operations ● Storage & Handling	8
Unit V	**Sales & Operations Planning (S&OP)** ● Contrast to Traditional & Siloed Approach to Distribution Planning ● Key drivers for Master Planning ● Conducting Pre-S&OP & S&OP Meetings ● Understanding S&OP ● Implementing S&OP	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	2	1	2	3	2
CO 2	2	1	3	3	1
CO 3	2	1	2	3	2
CO 4	2	1	2	3	1
CO 5	2	2	3	3	2

Action Based Component

- Role play and simulation replicating S&OP process
- Group case assignment on international transport planning
- Group case assignment on packaging, storage and handling in international trade

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission (assignments) (exam) day/week or length
			1	**2**	**3**	**4**	**5**	
1	Class Participation	5						
2	Attendance	5						

3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X				
6	CBT	10	X	X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Concepts in International Supply Chain Management – Archie D'Souza

2. Sales and Operations Planning, Shroff Publishing and Distributors, Co-published with APICS – Colleen Crum & George Palmatier

Reference books

1. International Logistics – The Management of International Trade Operations ~ Pierre David

2. Contemporary Logistics – Paul R Murphy, Jr & Donald F Wood

3. Logistical Management – The Integrated Supply Chain Process ~ Donald J Bowersox & David J Closs
4. The Handbook of Logistics & Distribution Management – Alan Ruston & others

Readings & Case Analysis

1. Harvard Business Review
2. MIT Sloan Management Review
3. ISTD. Indian society for Training & Development
4. Articles & literature from Cite HR website

Suggested Courses

1. Coursera courses on international trade, import/export, Sales and Operations Planning, Demand Planning, Supply Planning

SEMESTER/YEAR : **III SEM / II YEAR**
COURSE CODE : **25MBAS002**
TITLE OF THE COURSE : **TRANSPORTATION, INVENTORY & WAREHOUSE MANAGEMENT**
L: T: P: C : **3: 0: 2: 4**

Overview

Managing inventory effectively is key to the success of a manufacturing or sales/distribution organization. The cost of holding inventory is challenging to manage if the availability of materials is ensured at all times. This course covers the commonly used methodology for optimizing inventory levels. Transportation management and planning is an important function in supply chain management in order to meet delivery commitments on time and optimize transportation costs. Complex supply chains benefit from the use of optimization software tools to achieve these objectives. The course will provide hands-on experience on these tools. The execution aspects of transportation such as freight bill, packaging and various modes of transportation will be covered in the course. Distribution systems play a critical role in maintaining inventory and meeting the requirements of the customers. There are various types of warehouses and deciding their location and size is more of a science than art.

Course Objectives

1. Explain key terminology and concepts in inventory management and the mathematics of calculating various key parameters.
2. Use industry standard inventory control and planning tools.
3. Review the different modes of transportation, trade-offs and the resulting pricing and service implications
4. Analyze planning and management scenarios, business logics used in large scale planning of distribution and delivery
5. Identify and describe the challenges of setting up a distribution network, size and location of the warehouse and warehouse management practices

Course Outcomes

Upon successful completion of this course, the student will be able to-

1. Exhibit the fair understanding of concepts, theories, models and principles of transportation, inventory and warehousing management.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of transportation, inventory and warehousing management in real time problem solving
3. Demonstrate the ability of analyzing the real time problems using skill acquired by concepts, theories, models and principles of transportation, inventory and warehousing

management

4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of transportation, inventory and warehousing management in real-time problem-solving course
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of transportation, inventory and warehousing management in real-time problem-solving course

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case Analysis collaboratively with students involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of transportation, inventory and warehousing management practices by way of: ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

Units	Syllabus Details	Teaching Hours
Unit I	**Inventory Control and Management** ● Types and classification of inventory ● Importance of inventory management and challenges involved in it including conflicting goals ● Cost of carrying inventory, shelf-life challenges and storage costs ● Inventory related costs and economic order quantity ● Uncertainties in material supply from vendors	8
Unit II	**Inventory Performance Management** ● Variability in demand and supply and impact on inventory ● Popular inventory management methods ● ABC Classification ● Inventory Performance Management, Unit fill rate, line fill rate and Order fill rate	8

Unit III	**Transportation planning concepts** ● Terminology in transportation management such as freight bills, economics and pricing ● Various modes of transportation and applications ● Economics, costing, pricing in transportation ● Introduction to Sustainable Logistics	8
Unit IV	**Transportation planning in an enterprise** ● What tools are used in execution of transportation and distribution, transportation planning and management ● Considerations and trade-offs in transportation decisions ● Large scale distribution planning and the challenges involved ● Mathematical modelling of distribution and transportation scenarios	8
Unit V	**Distribution and Warehouse Management** ● Design of distribution network ● Size and location of warehouse ● Automation and mechanical handling system in warehouse ● Role of optimization in distribution planning and popular tools available	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	2	3	2	2	1
CO 2	2	3	2	2	1
CO 3	1	2	2	3	1
CO 4	2	3	2	2	2
CO 5	2	3	2	1	2

Action Based Component

- Mini Project on SC Network Design
- Group assignment on international trade
- Group assignment on Transportation planning

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X				
6	CBT	10	X	X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Spend Analysis: The Window into Strategic Sourcing by Kirit Pandit, H. Marmanis, J. Ross Publishing; 1st edition, 2008
2. Supply Chain Management: A Logistics Perspective by John J. Coyle, C. John Langley, Jr., Robert A. Novack, Brian J. Gibson, Cengage Publishing; 10th edition, 2017

Reference books

1. Procurement, Principles & Management by Peter Baily, David Farmer, Barry Crocker, David Jessop, David Jones; Pearson 11th Edition, 2015.
2. Essentials of Inventory Management by Max Muller; Amacom Publishing
3. Best Practice in Inventory Management by Tony Wild; published by Routledge

Readings & Case Analysis

1. Harvard Business Review

2. MIT Sloan Management Review
3. ISTD. Indian society for Training & Development
4. Articles & literature from Cite HR website

Suggested Courses

Coursera Courses

1. Supply Chain Logistics – Rutgers State University of New Jersey
2. Inventory Analytics - Rutgers State University of New Jersey

SEMESTER/YEAR : **III SEM / II YEAR**
COURSE CODE : **25MBAS003**
TITLE OF THE COURSE : **PROCUREMENT, FACTORY PLANNING & SCHEDULING**
L: T: P: C : **3: 0: 2: 4**

Overview

The course is designed to provide students an understanding of Sourcing, Procuring, Manufacturing planning and execution at scale. The complexity of planning the production of a mix of products within a given time horizon with limited machines and labour resources is a daunting task. Modern software systems provide these capabilities. The course will help students appreciate these complexities, use excel based planning to gain a hands-on understanding of planning an entire factory and re-plan based on changes in requirements. Students will understand various methods of load balancing, capacity utilization and planning orders under constraints and resolving conflicting scenarios to meet delivery commitments. Procurement of material plays a great role in the efficient functioning of a supply chain for any company. There are a fair number of complexities and challenges involved in carrying out procurement in time without overshooting budgets. The course covers the procedural aspects of procurement at a tactical level as well as strategic sourcing, vendor selection included in Strategic Relationship Management. The exercises in this course and assignments will be rooted in software tools used in the industry for SRM & Procurement. Vendor performance and review are critical for a company to make/alter sourcing decisions and the course will cover this aspect as well

Course Objectives

1. Identify the challenges in dealing with complex factory planning and scheduling scenarios; Understand factory lines, capacity and resource balancing and offloading of work orders
2. Explain the different types of manufacturing processes and their implication on planning and material movement
3. Develop an understanding of the mathematical methods and techniques used in factory planning software used in the industry today
4. Review purchase, procurement and strategic sourcing at a conceptual level; Understand the importance of inventory management including cost and service level
5. Describe the objectives of the purchase department in terms of minimizing Total Cost of Ownership, maintaining optimum inventory levels and managing vendor relations.

Course Outcomes

Upon successful completion of this course, the student will have reliably demonstrated ability to

1. Exhibit the fair understanding of concepts, theories, models and principles of procurement, factory planning and scheduling.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of procurement, factory planning and scheduling in real time problem solving
3. Demonstrate the ability of analyzing the real time problems using skill acquired by concepts, theories, models and principles of procurement, factory planning and scheduling
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of procurement, factory planning and scheduling in real-time problem-solving course
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of procurement, factory planning and scheduling in real-time problem-solving course

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case Analysis collaboratively with students involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Procurement and Supply Planning practices by way of: ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

Units	**Syllabus Details**	**Teaching Hours**
Unit I	**Procurement and Supplier lifecycle** Centralized vs decentralized procurement models; Strategic vs tactical sourcing; Role of procurement in competitive advantage – Market analysis and supplier segmentation, Category strategy development, spend analysis and opportunity identification; Supplier lifecycle – Onboarding, qualification, and certification of suppliers, Supplier Performance reviews and continuous improvement, Exit strategies and supplier consolidation.	8

Unit II	**Manufacturing Industries, Processes, Plant Layouts** ● Understanding of different types of industries and their manufacturing operations ● Discrete v/s continuous manufacturing processes and their implication on approach to planning ● Variety of plant layouts and implications on material movement and its efficiency	8
Unit III	**Hands-on factory planning and scheduling** ● Setup and use of industry standard factory planning, scheduling tools ● Handling of large-scale planning, data and analysis of planning scenarios ● Detailed scheduling of factory lines ● Resolution of infeasible planning scenarios based on prioritization with the help of tools	8
Unit IV	**Key goals of the purchase department** ● The importance of lowering the total cost of ownership and managing parts availability at minimum cost ● Selection and rationalization of vendors on an ongoing basis ● Managing delivery commitments to production divisions within the enterprise ● Introduction to Green and ethical procurement practices	8
Unit V	**Procurement operations and Vendor performance assessment** ● Criterion used in selection of vendors ● Day-to-day procedures used in procurement including order placement, management under budget and honoring supplier commitments ● Selection of vendors based on performance parameters and allocation of budgets to vendors ● Gathering data related to vendor performance ● Analysis of data on key parameters on prior agreed metrics	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	1	3	2	2	2
CO 2	2	3	2	2	1
CO	3	3	2	2	2

3					
CO 4	2	2	2	2	1
CO 5	2	3	2	2	2

Action Based Component

- Mini Project on procurement planning and scheduling
- ERP module of Procurement and P2P flow
- Group assignment on Selection of vendors and their performance assessment

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	**2**	**3**	**4**	**5**	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X				
6	CBT	10	X	X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Advanced Planning and Scheduling in Manufacturing and Supply Chains, by Yuri Mauergauz, 1st edition. 2016, SPRINGER.
2. Sourcing and Supply Chain Management by Robert B Handfield | Larry C Giunipero | James L Patterson | Robert M. Monczka published by Cengage

Reference books

1. Scheduling: Theory, Algorithms and Systems by Michael L. Pinedo,5th edition 2016, SPRINGER.
2. Procurement, Principles & Management by Peter Baily, David Farmer, Barry Crocker, David Jessop, David Jones, Pearson; 11th Edition, 2015.
3. Manufacturing Planning and Control Systems for Supply Chain Management by Vollmann Thomas published by McGraw-Hill

Readings & Case Analysis

1. Harvard Business Review
2. MIT Sloan Management Review
3. ISTD. Indian society for Training & Development
4. Articles & literature from Cite HR website

Suggested Courses

Coursera courses

1. Manufacturing Planning Control
2. Production planning, Workforce Management

SEMESTER/YEAR	**: III SEM / II YEAR**
COURSE CODE	**: 25MBAS004**
TITLE OF THE COURSE	**: DEMAND MANAGEMENT (DEMAND PLANNING & FORECASTING) & PROCUREMENT**
L: T: P: C	**: 3: 0: 2: 4**

Overview

This course is intended to enable students to understand the standard industry practices used in Demand Management. The course covers another very critical aspect of demand management, i.e., demand forecasting techniques and the mathematics behind them. Traditional methods of forecasting are covered including Static, Adaptive, and Auto-Regressive Techniques, Various measures of error and how to choose the right forecasting method is also covered. Students will learn current trends in forecasting methods including the use of Artificial Intelligence techniques such as Machine Learning, and Artificial Neural Network to augment statistical methods. The course will also provide an overview of Demand planning and techniques like Linear Programming that is used for resource allocation problems

Course Objectives

1. Learn the basics of demand management processes in large organizations
2. Understand mathematical underpinnings of forecasting
3. Understand the recent advances in forecasting
4. Learn hands on the use of demand forecasting software tools
5. Understand the Demand Planning Process

Course Outcomes

By the conclusion of this course, the student should be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of Demand management and procurement
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of Demand management and procurement in real time problem solving
3. Demonstrate the ability of analyzing the real time problems using skill acquired by concepts, theories, models and principles of Demand Management and procurement
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of Demand management and procurement in real-time problem-solving course
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of Demand management and procurement in real-time problem-solving course

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case Analysis collaboratively with students involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of demand management and procurement practices by way of: ● Mini projects ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

Units	**Syllabus Details**	**Teaching Hours**
Unit I	**Demand Forecasting in A Supply Chain Using Time Series** Understand the role of forecasting for both an enterprise and a supply chain. Introduction to Time Series Data. Components of time series data, measurement of trend, seasonality and cycles. Using moving averages and smoothing techniques for making a forecast. Holt 's and Winter 's smoothing model for forecasting. Various methods to measure Forecast Error. The Role of IT in Forecasting. Risk Management in Forecasting.	8
Unit II	**Multivariate Time Series Analysis and Forecasting** Vector error correction, vector autoregressive (VAR) models, their advantages and disadvantages, estimation and forecasting with VAR, Johansen Co-integration test on VAR, Granger causality test; Forecasting using regression models and their applications	8
Unit III	**Forecasting Using Regression Models** Demand forecasting using simple linear regression, Auto Regressive Moving Average [ARMA], Auto Regressive Integrated Moving Average [ARIMA], Seasonal Auto Regressive Moving Average [SARIMA], Auto Regressive Moving Average with Explanatory variable [ARMAX]	8
Unit IV	**Modern Demand Forecasting Using Machine Learning** Introduction to Machine Learning and Natural Language Processing. New methods in forecasting using Artificial Intelligence to supplement the mathematical forecast. Use of Neural Network for demand forecasting. Short term demand sensing and forecasting.	8

Unit V	**Enterprise Demand Planning** Demand management and forecasting in an enterprise. Implementing Demand Planning in an enterprise in practice. Various job functions related to demand management in a company. Annual demand planning process for the entire geography. Enterprise response to variability in demand. Understanding working with various parts of an organization in the demand planning process. Collaborating with sales, manufacturing and distribution. Manual override of the mathematical forecasts, rolling forecast revision and correction process	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	2	1	2	2	2
CO 2	1	3	2	2	1
CO 3	2	2	2	2	2
CO 4	2	2	1	1	2
CO 5	2	1	3	3	2

Action Based Component

- Mini Project on collaborative forecasting and planning
- Group assignment on calculating forecast using forecasting techniques and choosing the most accurate one
- Group case assignment on the impact of AI on forecasting

Course Assessment

#	**Description of Assessment Method**	**Weightage %**	**Learning Outcomes Assessed**					**Submission day/week (assignments) or length (exam)**
			1	**2**	**3**	**4**	**5**	
1	Class Participation	5						
2	Attendance	5						

3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X				
6	CBT	10	X	X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Demand Management Best Practices: Process, Principles, and Collaboration (J. Ross Publishing Integrated Business Management Series), Colleen Crum and George Palmatier, 2003

Reference books

1. Demand-Driven Forecasting: A Structured Approach to Forecasting, Charles W. Chase, 2013
2. Next Generation Demand Management: People, Process, Analytics, and Technology by Chrles W.Chase published by Wiley
3. Sales Forecasting Management: A Demand Management Approach by John T. Mentzer & Mark A. Moon published by Sage Publications

Readings & Case Analysis

1. Harvard Business Review
2. MIT Sloan Management Review
3. ISTD. Indian society for Training & Development
4. Articles & literature from Cite HR website

Suggested Courses

1. Coursera courses on forecasting and demand management

SEMESTER/YEAR : IV
SEM / II YEAR COURSE CODE :
25MBAS005
TITLE OF THE COURSE : SCM FOR BUSINESS IMPACT: SUPPLY CHAIN METRICS, SUPPLY CHAIN ANALYTICS AND PERFORMANCE MANAGEMENT
L: T: P: C : 3: 0: 2: 4

Overview

Efficient supply chain management can lead to impact on key business success parameters: profitability, customer satisfaction and revenue. As a result, Supply Chain Management is key not only for operational completeness but adds tangible value to any business. In order to achieve this, it is important to set appropriate KRAs and identify the right metrics, collect data and analyze the performance of the company on these parameters. This course will provide students an understanding of common metrics that are used in assessing the performance of a supply chain and tools used to analyze the data. Presentation of the right data, dashboard creation, measurement of performance and analytics at the correct level in the organization is covered in the course.

Course Objectives

1. Recognize and understand the importance of measurement of SC parameters.
2. Use data analytics tools to gather, present and make sense of historical data of the company supply chain
3. Understand platform businesses and retailing, including customer journey mapping
4. Create reports and perform data mining to extract useful information
5. Describe the characteristics of supply chains in various types of industries including e- commerce

Course Outcomes

Upon successful completion of this course, the student will have reliably demonstrated ability to:

1. Exhibit the fair understanding of concepts, theories, models and principles of SCM for Business Impact Course
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of SCM for Business Impact Course in real time problem solving
3. Demonstrate the ability of analyzing the real time problems using skill acquired by concepts, theories, models and principles of SCM for Business Impact course
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of SCM for Business Impact in real-time problem-solving course
5. Exhibit the ability of developing / redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of SCM for Business Impact in real-time problem-solving course

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Supply Chain practices by way of: ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

Units	**Syllabus Details**	**Teaching Hours**

Unit I	**Overview of Supply Chain Performance and its Business Impact** Basic understanding of supply chain performance parameters. Direct connection and dependency of revenue, profitability and customer satisfaction on these parameters and KRAs.	8
Unit II	**Overview of Analytics Tools** ● Overview of commonly used tools in Data Analytics: R, Tableau, Excel. ● Basics of building a data warehouse, ETL and reporting. ● Dashboard creation and insights using out-of-the-box reports. ● Deeper insights into supply chain metrics using customized reports and dashboards. ● Introduction to R programming and Tableau.	8
Unit III	**Reporting, dash boarding and data mining** ● Standard reports used by organizations for measurement and tracking of supply chain performance. ● Creation of these reports using R and excel. ● Analysis of historical data to discover trends and patterns that enable strategic decisions such as vendor selection. ● Use of simple data mining techniques in R. ● Strategic Challenges to retail and platform models and Future Trends	8
Unit IV	**Reporting, dash boarding and data mining** ● Standard reports used by organizations for measurement and tracking of supply chain performance. ● Creation of these reports using R and excel. ● Analysis of historical data to discover trends and patterns that enable strategic decisions such as vendor selection. ● Use of simple data mining techniques in R.	8
Unit V	**Strategic Supply Chain Management** ● Comparison of lean and agile supply chains. Advantages and disadvantages. ● Characteristics of supply chains in discrete, process, CPG, pharma and other industries. ● SCM in e-Commerce: challenges, opportunities. Difference between traditional SCM and e-Commerce ● SCM in omni-channel and retail industries	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	2	1	3	2	2
CO 2	1	3	2	2	3
CO 3	1	3	2	2	3
CO 4	1	3	2	2	3
CO 5	3	2	2	2	3

Action Based Component

- Mini Project on creating reports/dashboards for a dataset and obtaining insights
- Group assignment on one of the mathematical models
- Group assignment on e-commerce supply chain management

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission (assignments) (exam) day/week or length
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X				
6	CBT	10	X	X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Analytics in Operations/Supply Chain Management, 30 March 2016, Muthu Mathirajan, Chandrasekharan Rajendran, Sowmyanarayanan Sadagopan, Arunachalam Ravindran, Parasuram Balasubramanian

Reference books

1. The Applied Business Analytics Casebook: Applications in Supply Chain Management, Operations Management, and Operations Research, by Matthew J. Drake, 1st edition 2013, FT Pearson Press
2. Supply Chain Analytics: using data to optimize Supply Chain Processes by Peter W. Robertson published by Routledge
3. Supply Chain Analytics by T.A.S.Vijayaraghavan published by Wiley

Readings & Case Analysis

1. Harvard Business Review
2. MIT Sloan Management Review
3. ISTD. Indian society for Training & Development
4. Articles & literature from Cite HR website

Suggested Courses

Coursera Courses

1. Inventory Analytics - Rutgers State University of New Jersey

2. Operations and Supply Chain Decisions and Metrics – University of Illinois at Urbana- Champaign